different names for the same thing, and this means that, since all forms of mass produce gravity, so must all forms of energy. The Einstein field equation, when examined carefully, insists on it.] Now, if we have an enormously intense magnetic field—a field far more intense than ever encountered on Earth—then the field's intense energy will produce intense gravity, and that gravity will compress the field; it will hold the field lines together despite the pressure between them (Figure 7.1c). This was Melvin's discovery.

Wheeler's intuition told him that such "gravitationally bundled" field lines might be as unstable as a pencil standing on its tip: Push the pencil slightly, and gravity will make it fall. Compress the magnetic field lines slightly, and gravity might overwhelm their pressure, pulling them into implosion (Figure 7.1d). Implosion to what? Perhaps to form an infinitely long, cylindrical black hole; perhaps to form a *naked singularity* (a singularity without an enshrouding horizon).

It did not matter to Wheeler that magnetic fields in the real Universe are too weak for gravity to hold them together against explosion. Wheeler's quest was *not* to understand the Universe as it exists, but rather to understand the fundamental laws that govern the Universe. By posing idealized problems which push those laws to the extreme, he expected to gain new insights into the laws. In this spirit, he offered me my first gravitational research problem: Use the Einstein field equation to try to deduce whether Melvin's bundle of magnetic field lines will implode, and if so, to what.

For many months I struggled with this problem. The scene of the daytime struggle was the attic of Palmer Physical Laboratory in Princeton, where I shared a huge office with other physics students and we shared our problems with each other, in a camaraderie of verbal give-and-take. The nighttime struggle was in the tiny apartment, in a converted World War 11 army barracks, where I lived with my wife, Linda (an artist and mathematics student), our baby daughter, Kares, and our huge collie dog, Prince. Each day I carried the problem back and forth with me between army barracks and laboratory attic. Every few days I collared Wheeler for advice. I beat at the problem with pencil and paper; I beat at it with numerical calculations on a computer; I beat at it in long arguments at the blackboard with my fellow students; and gradually the truth became clear. Einstein's equation, pummeled, manipulated, and distorted by my beatings,